

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 02: CELL: THE BUILDING BLOCK OF LIFE

TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 Who first discovered and named 'cells' while examining thin cork slices in 1665?

- | | |
|---------------------------|--------------------|
| (A) Robert Brown | (B) Robert Hooke |
| (C) Anton van Leeuwenhoek | (D) Rudolf Virchow |

Q.2 Who first observed living, motile free cells (bacteria, protozoa) in pond water in 1674?

- | | |
|------------------|---------------------------|
| (A) Robert Hooke | (B) Anton van Leeuwenhoek |
| (C) Schleiden | (D) Schwann |

Q.3 Who formulated the aphorism 'Omnis cellula-e cellula' (all cells arise from pre-existing cells)?

- | | |
|--------------------|------------------|
| (A) Rudolf Virchow | (B) Robert Brown |
| (C) Camillo Golgi | (D) Purkinje |

Q.4 Which chemical compound makes up the bulk of the rigid plant cell wall?

- | | |
|-------------------|---------------|
| (A) Chitin | (B) Cellulose |
| (C) Peptidoglycan | (D) Glycogen |

Q.5 Which cell organelle is known as the 'powerhouse of the cell'?

- | | |
|---------------------|-------------------|
| (A) Ribosome | (B) Mitochondrion |
| (C) Golgi apparatus | (D) Lysosome |

Q.6 Which cellular organelle is responsible for synthesizing structural and enzymatic proteins?

- | | |
|--------------|----------------|
| (A) Lysosome | (B) Ribosome |
| (C) Vacuole | (D) Centrosome |

Q.7 Lysosomes are colloquially designated as:

- | | |
|-------------------------|------------------------------|
| (A) Kitchen of the cell | (B) Suicide bags of the cell |
| (C) Brain of the cell | (D) Powerhouse of the cell |

Q.8 The primary site of photosynthesis in green plant cells is the:

- | | |
|-----------------|-----------------|
| (A) Leucoplast | (B) Chromoplast |
| (C) Chloroplast | (D) Amyloplast |
-

Q.9 Which organelle plays a primary role in lipid synthesis and drug detoxification in liver cells?

- | | |
|---------------------|--|
| (A) Rough ER | (B) Smooth Endoplasmic Reticulum (SER) |
| (C) Golgi apparatus | (D) Nucleolus |
-

Q.10 Which organelle processes, packages, and sorts macromolecules received from the ER?

- | | |
|---------------------|----------------|
| (A) Golgi apparatus | (B) Peroxisome |
| (C) Vacuole | (D) Centriole |
-

Q.11 What happens to a plant cell placed in a hypertonic solution?

- | | |
|--------------------------|--|
| (A) It swells and bursts | (B) It undergoes plasmolysis (shrinks) |
| (C) It doubles in size | (D) It divides by mitosis |
-

Q.12 Which transport process describes the spontaneous movement of water across a semi-permeable membrane?

- | | |
|----------------------|-----------------|
| (A) Active transport | (B) Diffusion |
| (C) Osmosis | (D) Endocytosis |
-

Q.13 Prokaryotic cells lack which of the following cellular features?

- | | |
|------------------------------|---------------|
| (A) Plasma membrane | (B) Ribosomes |
| (C) A membrane-bound nucleus | (D) Cytoplasm |
-

Q.14 Which cellular organelle possesses its own DNA and ribosomes, enabling semi-autonomous replication?

- | | |
|-----------------------------------|---------------------|
| (A) Mitochondria and Chloroplasts | (B) Golgi apparatus |
| (C) Lysosomes | (D) Vacuoles |
-

Q.15 Which structure provides mechanical turgidity to mature plant cells, occupying up to 90% of cell volume?

- | | |
|---------------------|----------------|
| (A) Central Vacuole | (B) Centrosome |
| (C) Nucleolus | (D) Lysosome |
-

Q.16 Amoeba engulfs its food particles from the surrounding water by:

- | | |
|-----------------|--------------------------------|
| (A) Exocytosis | (B) Endocytosis (Phagocytosis) |
| (C) Plasmolysis | (D) Active secretion |
-

Q.17 What is chromatin chemically composed of?

- | | |
|--------------------------------|------------------------------|
| (A) RNA and lipids | (B) DNA and histone proteins |
| (C) Carbohydrates and minerals | (D) Phospholipids only |
-

Q.18 The membrane that encloses the plant central vacuole is called the:

- | | |
|-----------------|---------------|
| (A) Plasmalemma | (B) Tonoplast |
| (C) Periderm | (D) Pellicle |
-

Q.19 Which cellular structure is present in animal cells but absent in higher plant cells?

- | | |
|----------------------------|------------------|
| (A) Cell wall | (B) Chloroplast |
| (C) Centrosome / Centriole | (D) Mitochondria |
-

Q.20 Which stain is commonly used to stain plant cells like onion peel?

- (A) Safranin (B) Sudan III
(C) Benedict's reagent (D) Silver nitrate

Q.21 What is the full biological form of ATP?

- (A) Adenosine Triphosphate (B) Adenine Tetra-Phosphate
(C) Ammonium Tri-Phosphate (D) Adenosine Tartrate Phosphate

Q.22 What is the typical diameter of a human red blood cell (RBC)?

- (A) 7–8 μm (B) 70 μm
(C) 1 mm (D) 0.1 μm

Q.23 Which organelle is responsible for synthesizing ribosomes inside the eukaryotic nucleus?

- (A) Nucleolus (B) Nuclear pore
(C) Chromatin (D) Nuclear envelope

Q.24 Which plastid is responsible for storing starch grains in potato tubers?

- (A) Chloroplast (B) Chromoplast
(C) Amyloplast (Leucoplast) (D) Carotenoid

Q.25 What is the basic molecular composition of the fluid mosaic cell membrane?

- (A) Cellulose fibers only (B) Phospholipid bilayer with embedded proteins
(C) Solid wax layer (D) Pure glycogen mesh

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(B)	Robert Hooke observed dead honey-comb compartments in cork using an early microscope and coined the term 'cell'.
Q.2	(B)	Anton van Leeuwenhoek improved microscopic lenses and discovered living microorganisms.
Q.3	(A)	Rudolf Virchow expanded the cell theory in 1855 by stating that new cells arise from pre-existing cells by division.
Q.4	(B)	The primary structural component of plant cell walls is cellulose, a fibrous carbohydrate polymer.
Q.5	(B)	Mitochondria generate cellular energy in the form of ATP through aerobic cellular respiration.
Q.6	(B)	Ribosomes are the macromolecular protein-synthesizing factories found in all living cells.
Q.7	(B)	Lysosomes contain hydrolytic digestive enzymes that digest the cell itself if damaged, earning the name 'suicide bags'.
Q.8	(C)	Chloroplasts contain chlorophyll pigments and thylakoid membranes that capture light to perform photosynthesis.
Q.9	(B)	Smooth ER lacks ribosomes and specializes in lipid/steroid biosynthesis and hepatic detoxification.
Q.10	(A)	Camillo Golgi discovered the Golgi apparatus, which packages and modifies proteins for secretion.
Q.11	(B)	Water exits via exosmosis, causing the protoplast to shrink away from the cell wall (plasmolysis).
Q.12	(C)	Osmosis is the net movement of water molecules across a selectively permeable membrane along a concentration gradient.
Q.13	(C)	Prokaryotes lack a nuclear membrane; their genetic material lies naked in a region called the nucleoid.
Q.14	(A)	Both mitochondria and chloroplasts contain circular DNA and 70S ribosomes, reflecting endosymbiotic prokaryotic origin.
Q.15	(A)	The large central sap vacuole maintains hydrostatic turgor pressure against the plant cell wall.
Q.16	(B)	Plasma membrane flexibility enables Amoeba to form pseudopodia and engulf food via endocytosis.
Q.17	(B)	Chromatin consists of double-stranded DNA coiled around basic histone protein octamers.
Q.18	(B)	The selectively permeable membrane surrounding the large central plant vacuole is the tonoplast.
Q.19	(C)	Centrosomes and centrioles are animal cell organelles that organize spindle fibers during mitotic division.
Q.20	(A)	Safranin stains plant cell walls and nuclei bright pink/red for clear microscopic examination.
Q.21	(A)	ATP stands for Adenosine Triphosphate, the universal chemical energy storage molecule of cells.
Q.22	(A)	Human erythrocytes have an average diameter of approximately 7.2 to 7.8 micrometers (μm).
Q.23	(A)	The nucleolus is the dense subnuclear site of ribosomal RNA (rRNA) transcription and ribosome subunit assembly.
Q.24	(C)	Amyloplasts are specialized non-pigmented leucoplasts that synthesize and store starch in storage organs.
Q.25	(B)	Singer and Nicolson proposed the Fluid Mosaic Model: a phospholipid bilayer with embedded floating proteins.